

Qatar-1b

R-0237

Benchmark run · workflow W8-benchmark-deep · record deep-bench_Qatar-1_181005 · created 2026-07-17T04:30:54+00:00

BENCHMARK: INACCURATE

Results

Mid-Transit Time (Tmid)	2458396.8027 +/- 0.001 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.192 +/- 0.012
Transit depth (Rp/Rs)^2	3.69 +/- 0.47 [%]
Orbital Inclination (inc)	82.8 +/- 0.92
Airmass coefficient 1 (a1)	1.198 +/- 0.015
Airmass coefficient 2 (a2)	-0.1037 +/- 0.0058
Scatter in the residuals of the lightcurve fit is	1.22 %
Best Comparison Star	None
Optimal Aperture	3.32
Optimal Annulus	11.25
Transit Duration (day)	0.0651 +/- 0.0066

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.192 $\pm$ 0.012 vs 0.1463 (deviation 3.81 σ)
Duration fit vs published	0.0651 d vs 0.0692 d (deviation 0.62 σ)
Mid-time O–C	3.4 min
Depth SNR	16.0
Residual scatter	1.22 %

Light curve

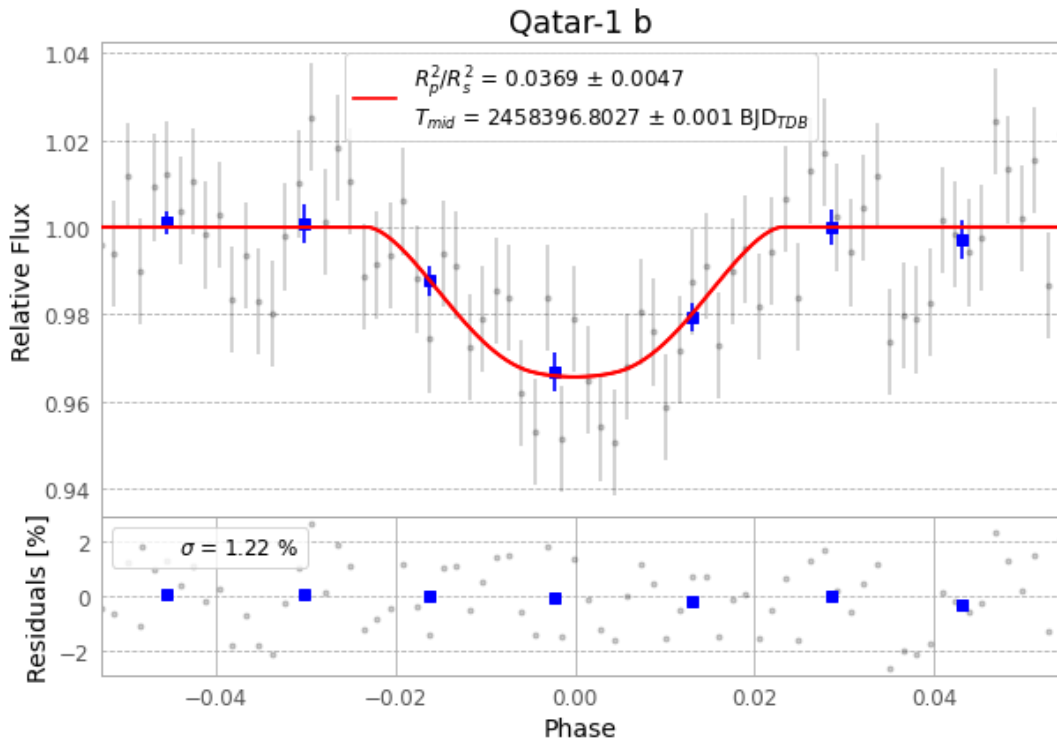


Figure 1 — FinalLightCurve_Qatar-1 b_2018-10-04.png (SHA-256 daf3a543dd2810cb...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [353, 412], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 fb992552447d7592...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 825eb7e

Data provenance

74 FITS frames (49 MB), Qatar-1181005052412.FITS → Qatar-1181005090312.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-Qatar-1-181005.log (da7a7b980010...), FinalLightCurve_Qatar-1 b_2018-10-04.png (daf3a543dd28...), FinalLightCurve_Qatar-1 b_2018-10-04.csv (9673fdc8d535...)

Compute resources

Wall time 120.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-17 04:30Z.